MATHS MANTHAN

• TUITION CLASSES •

e-mail: contact@mathsmanthan.co.in | Website:- mathsmanthan.co.in

Subject: Mathematics | **CBSE CLASS 10 – WEEKLY TEST 04** | **Time:** 45 Minutes

Class: Class X (CBSE) | **Chapter 10: Circles** | **MM:** 20

Document Type: Official Step-by-Step Marking Scheme | **Published for:** mathsmanthan.co.in

CBSE STEP-MARKING GUIDELINES & EVALUATION PROTOCOL:

1. Full marks require explicit formula declaration, correct algebraic substitution, and final boxed answer with proper physical units.
2. In alternative (**OR**) questions, step marking applies equally to both options.
3. Partial marks are allocated per step as indicated in brackets: **[0.5M, 1M, 2M]**. Alternative mathematically valid methods are fully credited.

Official Evaluation Key | Maths Manthan Meerut

SECTION A: Multiple Choice & Assertion-Reason

MATHS MANTHAN | (5 × 1 = 5 Marks)

Q1. A circle touches all four sides of a quadrilateral $PQRS$ with $PQ = 11$ cm, $QR = 8$ cm, and $RS = 7$ cm. The length of SP is: [1]

(A) 9 cm (B) 10 cm (C) 12 cm (D) 14 cm

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (B) 10 cm

Justification: Since opposite sides sum equally in a circumscribed quadrilateral: $PQ + RS = QR + SP \implies 11 + 7 = 8 + SP \implies SP = 18 - 8 = \boxed{\mathbf{10}\text{ cm}}$.

Q2. If the angle between two tangents drawn from a point P to a circle of radius a and centre O is 90°, then OP is equal to: [1]

(A) a (B) $a\sqrt{2}$ (C) $a/\sqrt{2}$ (D) $2a$

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (B) $a\sqrt{2}$

Justification: With tangent angle 90°, $\angle OPA = 45°$. In right $\triangle OAP$: $\sin 45° = \frac{a}{OP} \implies \frac{1}{\sqrt{2}} = \frac{a}{OP} \implies \boxed{OP = \mathbf{a\sqrt{2}}}$.

Q3. In two concentric circles of radii 10 cm and 8 cm, the length of a chord of the outer circle which touches the inner circle is: [1]

(A) 6 cm (B) 12 cm (C) 16 cm (D) 18 cm

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (B) 12 cm

Justification: Half-chord length $= \sqrt{10^2 - 8^2} = \sqrt{36} = 6$ cm $\implies$ Chord length $= 2 \times 6 = \boxed{\mathbf{12}\text{ cm}}$.

Q4. Tangents TP and TQ are drawn from external point T to a circle with centre O. If $\angle OPT = 90°$ and $\angle PTQ = 40°$, then $\angle OPQ$ is: [1]

(A) $15°$ (B) $20°$ (C) $25°$ (D) $30°$

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (B) $20°$

Justification: Since $TP = TQ$, $\angle TPQ = \frac{180° - 40°}{2} = 70°$. Hence $\angle OPQ = \angle OPT - \angle TPQ = 90° - 70° = \boxed{20°}$.

Q5. **Assertion (A):** The tangent at any point of a circle is perpendicular to the radius through the point of contact.

Reason (R): The shortest distance of a point from a given straight line is the perpendicular distance. [1]

(A) Both (A) and (R) are true and (R) is the correct explanation of (A).

(B) Both (A) and (R) are true but (R) is not the correct explanation of (A).

(C) (A) is true but (R) is false.

(D) (A) is false but (R) is true.

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (A)

Justification: For any other point Q on the tangent outside the circle, $OQ > r = OP$, so OP is the shortest distance from O to the tangent line. By Reason (R), OP must be perpendicular to the tangent. Both are true and (R) explains (A).

SECTION B: Very Short Answer Type

MATHS MANTHAN | $(1 \times 2 = 2$ Marks)

Q6. From an external point P, two tangents PA and PB are drawn to a circle with centre O. If $\angle AOB = 130°$, find the measure of $\angle APB$. [2]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

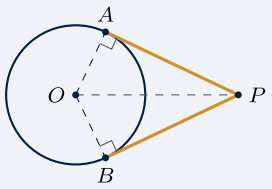
Step 1: Perpendicularity Property [1 Mark]

By Theorem 10.1, $\angle OAP = 90°$ and $\angle OBP = 90°$.

Step 2: Angle Sum in Quadrilateral $OAPB$ [1 Mark]

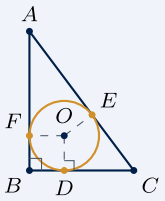
$\angle OAP + \angle AOB + \angle OBP + \angle APB = 360° \implies 180° + 130° + \angle APB = 360°$.

$$\therefore \angle APB = 180° - 130° = \boxed{50°}$$



Q7. If a, b, c are the sides of a right triangle where c is the hypotenuse, prove that the radius r of the circle which touches the sides of the triangle is $r = \frac{a+b-c}{2}$. **[3]**

▶ STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 3 Marks]



Let $\triangle ABC$ be right-angled at B with $AB = a, BC = b, AC = c$, and incircle touches BC, CA, AB at D, E, F.

Step 1: Identifying the Square $ODBF$ [1 Mark]

Since $OD \perp BC$, $OF \perp AB$, $\angle B = 90°$, and $OD = OF = r$, quadrilateral $ODBF$ is a square, so $BD = BF = r$.

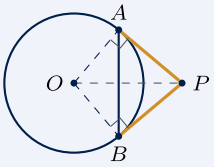
Step 2: Using Equal Tangents from A and C [2 Marks]

$AF = AE = a - r$ and $CD = CE = b - r$. Since $AC = AE + CE$:

$$c = (a-r)+(b-r) \implies 2r = a+b-c \implies \boxed{r = \frac{\mathbf{a+b-c}}{\mathbf{2}} \quad \text{[Proved]}}$$

Q8. From an external point P, two tangents PA and PB are drawn to a circle with centre O. If chord AB subtends an angle of $100°$ at the centre, find $\angle APB$ and $\angle PAB$. **[3]**

▶ STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 3 Marks]



Step 1: Finding $\angle APB$ [1.5 Marks]

Since $\angle OAP = \angle OBP = 90°$, angle sum in quad $OAPB$ gives:

$$\angle APB = 180° - \angle AOB = 180° - 100° = \mathbf{80°}$$

Step 2: Finding $\angle PAB$ [1.5 Marks]

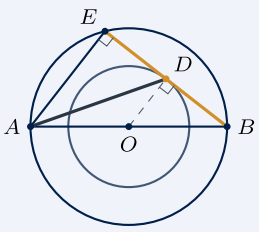
In $\triangle PAB$, $PA = PB$ (Theorem 10.2), so $\angle PAB = \angle PBA$:

$$\angle PAB = \frac{180° - \angle APB}{2} = \frac{180° - 80°}{2} = \mathbf{50°}$$

$$\boxed{\angle APB = \mathbf{80°}, \quad \angle PAB = \mathbf{50°}}$$

Q9. Two concentric circles have radii 13 cm and 8 cm. AB is a diameter of the larger circle and BD is a tangent to the smaller circle touching it at D and meeting the larger circle at E. Find the length of AD. **[3]**

▶ STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 3 Marks]



Step 1: Properties at D and E [1.5 Marks]

$OD \perp BE$ (radius $\perp$ tangent) $\implies$ D bisects chord BE ($BD = DE$). Also AB is a diameter, so $\angle AEB = 90°$ (angle in semicircle). Since O and D are midpoints of AB and BE, $OD = \frac{1}{2}AE \implies AE = 2 \times 8 = 16$ cm.

Step 2: Pythagoras Theorem in $\triangle ODB$ and $\triangle AED$ [1.5 Marks]

$DE = BD = \sqrt{OB^2 - OD^2} = \sqrt{13^2 - 8^2} = \sqrt{105}$ cm. In right $\triangle AED$:

$$AD = \sqrt{AE^2 + DE^2} = \sqrt{16^2 + 105} = \sqrt{256 + 105} = \sqrt{361} = \boxed{\mathbf{19}\text{ cm}}$$

Q10. Satellite Horizon Line-of-Sight Coverage

A communications satellite orbiting at point S maintains line-of-sight contact with the Earth's horizon at tangent stations A and B. Earth is modeled as a sphere whose cross-section is a circle of radius $R = 6400$ km with centre C. The distance from the satellite to Earth's centre is $SC = 10000$ km.

(i) State why triangle $\triangle CAS$ is a right-angled triangle. [1]

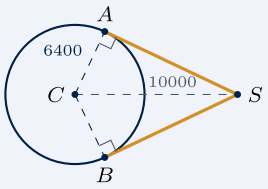
(ii) Calculate the distance SA from the satellite to horizon station A. [1]

(iii) Determine the cosine of angle $\angle ACS$, OR find the distance of the satellite above the Earth's surface. [2]

▶ STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 4 Marks]

(i) Why $\triangle CAS$ is Right-Angled: [1 Mark]

SA is tangent to the Earth's horizon circle at A and CA is the radius; by Theorem 10.1, $CA \perp SA \implies \boxed{\angle \mathbf{CAS = 90^\circ}}$.



(ii) Line-of-Sight Distance SA: [1 Mark]

$SA = \sqrt{SC^2 - CA^2} = \sqrt{10000^2 - 6400^2} = 400\sqrt{25^2 - 16^2} = 400\sqrt{369} = \boxed{\mathbf{1200\sqrt{41}}\text{ km} \approx \mathbf{7683.75}\text{ km}}$.

(iii) Main Choice — Value of $\cos(\angle ACS)$: [2 Marks]

In right $\triangle CAS$ ($\angle CAS = 90^\circ$): $\cos(\angle ACS) = \frac{\text{Adjacent } (CA)}{\text{Hypotenuse } (SC)} = \frac{6400}{10000} = \boxed{\mathbf{\frac{16}{25} = 0.64}}$.

— OR — (Alternative Choice — Altitude Above Earth's Surface): [2 Marks]

Altitude above the surface $= SC - R = 10000\text{ km} - 6400\text{ km} = \boxed{\mathbf{3600}\text{ km}}$.

★★★ END OF SOLUTIONS · MATHS MANTHAN MEERUT ★★★